Bengali Word Sense Disambiguation Using Supervised approach (BERT)

by

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Capstone project report (CSE499) submitted in partial fulfillment of the requirements for the degree of

Bachelor of Science in Computer Science and Engineering

Under the supervision of

Raihan Kibria



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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SPRING 2025

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DECLARATION

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We declare that this capstone project report entitled *Bengali Word Sense Disambiguation Using Supervised approach (BERT)* is the result of our own work except as cited in the references. The capstone project report has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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Ariful Islam Bhoiyan
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Date: May 12, 2025

Dedicated To my beloved
Elder sister,
whose unwavering support, endless encouragement, and quiet strength have been a guiding light in my life and throughout this journey. Your faith in me has been my greatest motivation. This work is as much yours as it is mine.

– Ariful Islam Bhoiyan

ABSTRACT

Word Sense Disambiguation (WSD) is a pivotal challenge in Natural Language Processing (NLP) for morphologically complex languages like Bengali, spoken by over 250 million people. Polysemous Bengali words, with multiple meanings based on context, create significant obstacles for applications such as machine translation and information retrieval. This study introduces a supervised WSD framework utilizing transformer-based BERT models to tackle these issues. We present a curated Bengali WSD dataset with contextually labeled examples, encompassing a variety of polysemous words and their senses. Three BERT-based models—csebuetnlp/BanglaBERT, sagorsarker/bangla-bert-base, and a multilingual variant—are evaluated within a consistent pipeline. The dataset is divided into 80% for training and 20% for testing, with preprocessing to address Bengali's morphological intricacies. Model performance is assessed using accuracy, precision, recall, F1-score, and confusion matrix analysis. The csebuetnlp/BanglaBERT model achieved the highest performance, with 91.51% accuracy, 92.43% precision, 91.51% recall, and 91.84% F1-score, surpassing sagorsarker bangla-bert-base (89.84% accuracy) and other models. Class-wise analysis highlights robust performance in semantically distinct categories but reveals challenges with nuanced or underrepresented senses. We recommend csebuetnlp/BanglaBERT as the optimal model for Bengali WSD due to its superior accuracy, reliable metrics, stable training, and strong generalization across diverse contexts. This framework and dataset provide a robust foundation for advancing Bengali NLP in low-resource, high-complexity settings.

Keywords: Word Sense Disambiguation, Bengali NLP, Transformer Models, BanglaBERT, Polysemy, Supervised Learning, Contextual Embeddings

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Chapter 1

Introduction

Natural Language Processing (NLP) involves the interaction between computers and human languages, with Word Sense Disambiguation (WSD) being a fundamental problem. It involves identifying the correct sense of a word based on its context in a sentence, crucial for machine translation, information retrieval, question answering, and other downstream applications. Accurately determining these meanings is essential for accurate translation and comprehension.

In the context of Bengali—a morphologically rich and syntactically complex language spoken by over 250 million people—WSD remains a relatively underexplored area. Many Bengali words are polysemous, meaning they have multiple meanings that vary depending on context. For example:

(mool) can mean ”price”, ”root”, or ”main”.

(adhar) can mean ”foundation”, ”support”, or a ”dark-light ambiance”.

Incorrectly interpreting such words may cause serious misunderstandings in translation, dialogue systems, or search engines.

Traditional WSD approaches—such as dictionary-based, knowledge-based, or supervised classification with hand-engineered features—have limited success in Bengali due to:

Lack of comprehensive sense inventories like WordNet (available for English).

Scarcity of annotated corpora.

High inflection and derivation in Bengali morphology.

Variability in syntactic structures across dialects and registers.

Recent advances in transformer-based models, particularly BERT, have revolutionized Bengali NLP by enabling models to learn contextual embeddings, making

them ideal for tasks like Word Sight Decoding. Variants like sagorsarker/bangla-bert-base, csebuetnlp/banglabert, and multilingual models have opened new possibilities.

However, applying BERT-based models to Bengali WSD presents unique challenges:

A lack of benchmark WSD datasets with contextually labeled examples.

Need for domain-general examples to ensure models can generalize across topics.

Performance evaluation across different transformer models in a consistent pipeline.

This study proposes a supervised WSD framework for Bengali that leverages BERT-based transformer architectures. It introduces a curated Bengali WSD dataset and compares the performance of three models—Bangla-BERT, BanglaBERT. This task using accuracy, F1-score, and confusion matrix-based analysis.

1.1 Problem statement

Despite the growing availability of transformer-based models for Bengali, the task of Word Sense Disambiguation remains largely unsolved. Most existing research is either limited to English or restricted to shallow approaches in Bengali (e.g., rule-based or statistical). A lack of large-scale, annotated datasets has also inhibited experimentation with deep learning methods for Bengali WSD.

The central research problem addressed in this study is:

"How effective are transformer-based BERT models in disambiguating the correct sense of polysemous Bengali words in a supervised setting?"

This problem encapsulates several sub-problems:

Can we construct a labeled WSD dataset for Bengali that covers diverse words and senses?

Which BERT-based models perform best on Bengali WSD?

How do these models handle contextual ambiguity, especially with similar senses?

What are the limitations of current models in low-resource, high-complexity language settings like.

1.2 Aims, objectives and Motivation

1.2.1 Aims

Building a supervised learning-based Word Sense Disambiguation (WSD) system for the Bengali language utilizing transformer-based models—more especially, BERT and its variants—is the main goal of this study. The project builds a high-quality, annotated WSD dataset and benchmarks cutting-edge models in an effort to close the gap in Bengali NLP resources. Enabling machines to understand the complex meaning of Bengali words in a variety of settings is the ultimate objective, since it is essential for several downstream applications including question answering, information retrieval, and machine translation. To realize this overarching objective, the following specific aims are outlined:

- **Design and Develop a Supervised WSD System:** Using refined transformer models, formulate Bengali WSD as a multi-class classification issue and develop a system that can correctly disambiguate polysemous words based on sentence-level context.
- **Construct a Rich Annotated Bengali WSD Dataset:** At least 122 polysemous Bengali words, each linked to three to five different senses and 40 to 50 contextually varied phrases per sense, should be included in a manually validated dataset. By using samples from books, news stories, and conversational texts, you may guarantee linguistic diversity. Arrange the data in machine-readable forms that are appropriate for contemporary NLP pipelines, such as CSV or JSON.
- **Utilize Pre-trained Bengali Transformer Models:** Employ and fine-tune three prominent pre-trained transformer models — `sagorsarker/bangla-bert-base`, `csebuetnlp/banglabert` — to evaluate their effectiveness in capturing semantic nuances in Bengali for WSD tasks. Leverage the Hugging Face Transformers library and PyTorch as the technical backbone.
- **Incorporate Strategies for Class Imbalance:** The intrinsic class imbalance issue brought on by different sense frequency distributions in natural language corpora may be addressed by using strategies like weighted cross-entropy loss and stratified sampling.

- **Implement Comprehensive Evaluation Metrics:** Utilize the Accuracy, Precision, Recall, F1-Score, and confusion matrices to assess each model. Monitor learning curves (accuracy and loss per epoch) to spot problems with generalization and overfitting. Analyze qualitative errors to find systemic flaws in the model.

- **Enable Model Comparison and Insightful Analysis:** Examine how well the chosen models perform in various language categories and syntactic complexity levels. Examine their capacity to generalize across unclear structures or unrecognized word senses, and note the advantages and disadvantages of each model.

- **Advance Bengali NLP and Support Downstream Tasks:** Show how a dependable Bengali WSD system may reduce semantic ambiguity and improve downstream tasks like sentiment analysis, chatbot interactions, and Bengali-to-English machine translation.

- **Contribute to Open Research and Reproducibility:** Make the training scripts, assessment benchmarks, and annotated dataset openly accessible. Give thorough documentation so that future scholars can expand on this work, especially for other South Asian languages with limited resources.

1.2.2 Objective

To achieve the aim of this research, the following objectives are set:

1. **Data Construction**

    - Collect polysemous Bengali words and define their distinct senses.
    - Manually annotate at least 40-122 words, each with 4-5 senses, and 40-50 context-rich sentences per sense.
    - Store the dataset in a structured format (e.g., JSON) suitable for transformer input pipelines.

2. **Model Selection and Fine-tuning**

    - Choose transformer models pretrained on Bengali:

  - `sagorsarker/bangla-bert-base`
  - `csebuetnlp/banglabert`
- Fine-tune each model using the training set while applying stratified validation/test splits.
- Apply weighted loss functions to address class imbalance during training.

3. **Evaluation**
   - Evaluate model performance using Accuracy, Precision, Recall, and F1-Score.
   - Generate and interpret confusion matrices to analyze sense-level misclassifications.
   - Visualize training curves (loss and accuracy) and identify signs of overfitting or underfitting.
4. **Comparison and Analysis**
   - Compare the three models across performance metrics and analyze their respective strengths and weaknesses.
   - Conduct error analysis to identify common misclassification patterns or systematic model biases.
   - Determine if any model shows better generalization across different sense categories.
5. **Documentation and Publication**
   - Document the entire dataset creation process, annotation guidelines, and modeling configurations.
   - Make the dataset and code publicly available in a reproducible format to support future research.

1.2.3 Motivation

The motivation behind this research is multifaceted:

- **Lack of Bengali NLP Resources:**
  The majority of Word Sense Disambiguation (WSD) systems are designed for European languages, including English. Because Bengali is a low-resource language, benchmarking and model improvement are challenging due to the lack of datasets and resources for supervised WSD tasks.

- **Utility in Downstream Tasks:**
  For idiomatic idioms and domain-specific terminology in particular, accurate sense disambiguation is essential to improving downstream applications like machine translation. Additionally, it enhances Bengali voice assistant and chatbot interactions by giving more accurate and contextually relevant answers.

- **Transformer Potential:**
  Transformer-based models, such as BERT, are intrinsically ideal for WSD owing to their extensive contextual awareness of language. Their demonstrated success in other languages drives their applicability in Bengali.

- **Linguistic Richness of Bengali:**
  Bengali is renowned for its extensive use of metaphors, lyrical phrases, and colloquial variants. Modeling these subtleties can improve semantic comprehension and cultural representation in NLP systems.

- **Research Contribution:**
  This research builds a custom-annotated dataset and conducts a comparative evaluation of BERT-based models for Bengali WSD. The outcomes can benefit not only Bengali NLP but also similar low-resource Indo-Aryan languages.

1.3 Project Specification

This project is titled *Supervised Bengali Word Sense Disambiguation Using BERT-based Models* and is framed as a supervised classification task. The input consists of

Bengali sentences containing polysemous words, and the output is a predicted sense label. The key technical components include:

- A bespoke Bengali WSD dataset including at least 122 polysemous words, each with 3-5 senses and around 50 annotated example sentences per sense.
- A modeling pipeline created with the Hugging Face Transformers library and a PyTorch backend.
- Support for weighted cross-entropy loss to mitigate the effects of class imbalance during training.
- Evaluation through confusion matrices, classification reports, and training plots to analyze performance and model behavior.
- Cross-model evaluation using three transformer-based models: `sagorsarker/bangla-bert-base`, `csebuetnlp/ banglabert`.

Ultimately, this project aims to establish a reliable benchmark and dataset for Bengali WSD, providing a foundational resource for future research and contributing to the advancement of semantic understanding in Bengali language processing.

- Give a clear set of design specifications for the project. The design specifications should be clear concise statements with a specific metric and an appropriate value.
- The specifications should provide a specific measure of the success of the final design in meeting the need and constraints associated with the design problem.
- Problem Requirements Specifications is a dynamic process. Although it is desirable to freeze a set of requirements permanently, it is rarely possible. Requirements are likely to evolve through an iterative process that involves communication between customers specifying the need and the technical community. The impact of proposed requirements must be evaluated to ensure that the initial intent of the requirements baseline is maintained or that changes to the intent are understood and accepted by the customer.

Chapter 2

Literature Review

Word Sense Disambiguation (WSD) is a fundamental job in Natural Language Processing (NLP) that aims to detect the right meaning of a word based on its context. In recent years, deep learning (DL) approaches, notably transformer-based models, have made substantial progress in increasing WSD accuracy across a wide range of languages, including Bengali. Several research have investigated numerous methodologies, each with merits and weaknesses.

In 2022, Das et al. introduced a dataset specifically designed for evaluating Bengali WSD techniques, which includes 100 polysemous words, each with 3 to 4 distinct senses. Although the dataset is small and narrowly scoped, it offers valuable insights into Bengali language-specific challenges in WSD and provides a foundation for testing and benchmarking future models. Despite its limited scope, this dataset is instrumental in evaluating initial methods and laying the groundwork for more complex models [1].

Another significant contribution to Bengali WSD was made by Pal and Saha (2019), who proposed an unsupervised methodology that combined Principal Component Analysis (PCA) and Bengali WordNet to address word sense disambiguation. Their method achieved a reasonable accuracy of 61%. However, this approach did not use supervised transformer models, which have shown superior results in recent years. The reliance on older machine learning techniques, such as PCA, is a limitation when compared to transformer models like BERT, which have been more successful in capturing contextual word meanings in large datasets [2].

In 2023, Saidi and Jarray presented a novel hybrid approach that combined BERT and Convolutional Neural Networks (CNN) for Arabic WSD. They proposed the BERT-CNN model and evaluated it against several variations, including

ArabicBERT-CNN and AraBERT-CNN. Their models achieved impressive performance, with accuracy rates of 96.42% for BERT-CNN, showing the potential of combining transformer models with CNNs to enhance WSD accuracy. This approach outperformed traditional methods and highlighted the benefits of combining pre-trained language models like BERT with domain-specific architectures such as CNNs [3].

Similarly, in 2024, Holanda do Nascimento et al. focused on WSD for the Portuguese language using BERTimbau, a BERT-based model tailored for Portuguese. Their model achieved an accuracy of 84%, further demonstrating the effectiveness of transformer models in multilingual WSD tasks. Despite the strong results, the BERTimbau model still faced challenges in handling polysemy in resource-scarce languages and contexts that require more extensive fine-tuning [4].

In the same year, Kumari and Lobiyal explored Hindi WSD using distributed word representations such as Word2Vec (Skip-Gram) and Probabilistic Latent Semantic Analysis (PLSA). Their study reported an accuracy of 74.12% for the Word2Vec model and 72% for PLSA. However, their study lacked the incorporation of modern transformer-based models, which have proven to be more effective in capturing the complexities of polysemy and context. The absence of transformers in their approach indicates a gap in utilizing cutting-edge models for Hindi WSD tasks [5].

Lastly, Sarker (2020) proposed the Bangla BERT model, which was one of the first transformer-based models specifically designed for Bengali. This model laid the foundation for future research and advancements in Bengali NLP. Although the model was limited by the size and scope of its training data, it demonstrated the viability of transformer models for resource-poor languages. As Bengali NLP continues to grow, models like Bangla BERT are expected to play a crucial role in improving WSD tasks and addressing challenges related to linguistic diversity [6].

2.1 Materials and Methods:

A supervised Bengali Word Sense Disambiguation (WSD) framework was developed using deep learning models based on transformer architectures. The dataset was custom-curated and consisted of multiple Bengali target words, each associated with several distinct senses. For each sense, a set of example sentences was

manually created to ensure semantic diversity and clarity. On average, each word had 4–5 sense labels and 40–50 sentences per sense, ensuring a balanced dataset suitable for supervised training.

The dataset was preprocessed and formatted in JSON, containing fields such as the target word, sense label, and the sentence context. Stratified splitting was used to divide the dataset into training and testing sets while preserving the proportional distribution of sense classes for each word.

Three transformer-based models—BanglaBERT and Bangla-BERT-Base fine-tuned on this dataset. Each sentence was tokenized using the corresponding tokenizer of the model. Padding and truncation were applied to maintain a uniform sequence length. A classification head was added to the final layer of the transformer output to predict the correct sense of the word within its context.

The training process used cross-entropy loss with class weights to address class imbalance. The models were trained using the AdamW optimizer with a learning rate of 5e-5. Early stopping was applied based on validation loss to prevent overfitting. Transfer learning was employed by initializing models with pre-trained weights before fine-tuning them on the WSD task.

Evaluation of the models was conducted using standard classification metrics including accuracy, precision, recall, F1-score, and macro-averaged AUC-ROC. Confusion matrices were generated for each word to analyze sense-specific performance. A 5-fold cross-validation approach was used to ensure generalizability and stability of the model performance.

Misclassified samples were manually analyzed to identify patterns such as overlapping senses, ambiguous sentence structures, or insufficient context, which helped to better understand the limitations and guide potential future improvements in data or model design.

2.2 Result:

Deep learning models have shown significant promise in the area of Word Sense Disambiguation (WSD), particularly for Bengali, using both supervised and unsupervised methodologies. The proposed models, which integrated BERT-based transformers, demonstrated high accuracy and strong performance across diverse WSD tasks. Das et al. (2022), for instance, introduced a dataset for evaluating

Bengali WSD techniques and achieved a solid baseline accuracy despite the dataset being small. The dataset's polysemy scope and diverse contexts enabled a robust test environment, although it was limited by its size, which suggests the need for larger, more comprehensive datasets in the future.

Pal and Saha (2019) explored unsupervised methods combined with PCA and Bengali WordNet. The accuracy of their method was 61%, which, although a decent starting point, fell short when compared to transformer-based models. This highlights the gap in utilizing more advanced models such as BERT, which have demonstrated much better performance in the field. The methodology did not fully exploit the potential of pre-trained transformer models, which would have allowed for a more nuanced understanding of word senses in various contexts.

In a parallel study on Arabic WSD, Saidi and Jarray (2023) achieved remarkable results by stacking BERT with CNNs, reaching an accuracy of 96.42% with their BERT-CNN model. This hybrid approach significantly outperformed traditional methods and set a high standard for future multilingual WSD models. The study indicated that combining BERT's contextual understanding with CNN's ability to extract features from local contexts was an effective strategy, particularly for languages with complex morphology, such as Arabic.

Holanda do Nascimento et al. (2024), focusing on Portuguese, employed the BERTimbau model and achieved an accuracy of 84%. Their work demonstrated the strong applicability of transformer models in WSD, though the results suggest that there is still room for improvement in handling polysemy in languages with rich syntactic and semantic structures.

In Hindi, Kumari and Lobiyal (2022) evaluated Word2Vec and PLSA models for WSD, achieving a maximum accuracy of 74.12%. The study highlighted the limitations of using traditional word embedding models without the integration of advanced techniques like transformers. Their results also suggested that while Word2Vec was effective for simpler WSD tasks, it was less robust in distinguishing between similar senses of words compared to more sophisticated approaches like BERT.

Finally, Sarker (2020) introduced the Bangla BERT model, marking a milestone in Bengali NLP. Although the model's development was limited by the available training data, it served as an essential building block for future research and ad-

vancements in Bengali WSD. The model set the groundwork for further exploration into the potential of transformer models for resource-poor languages like Bengali.

The combination of advanced transformer models and better-structured datasets will continue to drive improvements in Bengali WSD. While current models have shown promising results, challenges such as handling fine-grained senses, improving accuracy for low-resource languages, and addressing data privacy remain areas for further research. As demonstrated across the studies, privacy-preserving techniques like federated learning and secure data aggregation, when applied to WSD, offer a path for ensuring the confidentiality of training data while still achieving strong model performance.

2.3 Comparison Literature accuracy table:

2.4 Discussion:

The exploration of privacy-preserving techniques within deep learning, particularly for Word Sense Disambiguation (WSD) in Bengali, has seen significant progress in recent years. Traditionally, machine learning models have been reliant on centralized storage, which raises concerns about the security and privacy of sensitive linguistic data. In response, federated learning (FL) and transformer-based models have emerged as promising solutions. Federated learning enables decentralized model training, which ensures that sensitive data, such as word senses and their context in training datasets, never needs to leave the local devices. This approach aligns with the growing need for secure and privacy-preserving techniques in the field of natural language processing (NLP), especially when dealing with sensitive datasets in resource-constrained languages.

Recent studies have integrated deep learning models, particularly BERT-based transformers, with federated learning to tackle complex linguistic tasks such as WSD. Transformer models, due to their self-attention mechanism, are well-suited for tasks like WSD, where understanding long-range dependencies and contextual information is essential. For Bengali WSD tasks, transformer-based models like sagorsarker/bangla-bert-base and csebuetnlp/banglabert have proven effective at capturing the nuances of word meanings across different contexts. These models have outperformed traditional methods like Word2Vec and Latent Semantic

Analysis (LSA), demonstrating a significant improvement in accuracy and context sensitivity. Despite these advancements, their application in federated learning presents new challenges.

Privacy preservation remains a critical concern in federated learning environments, especially when sensitive linguistic data, such as user-generated content and domain-specific usage, are involved. In the case of Bengali WSD, model updates transmitted between clients and servers can still potentially leak indirect information about the data. To mitigate such risks, privacy-preserving techniques such as differential privacy, homomorphic encryption, and secure multi-party computation have been integrated into federated learning frameworks. These techniques aim to ensure that no sensitive information is exposed during the model training process. However, challenges still remain in balancing the trade-off between privacy and accuracy.

A key challenge in applying federated learning to Bengali WSD is data heterogeneity. Medical image datasets have shown similar issues with varying data quality and annotation standards across institutions. In the case of Bengali, different regional and domain-specific variations in word senses and contexts can create model biases, leading to reduced generalization. This issue is particularly pronounced in languages with rich morphology like Bengali, where word senses can be highly nuanced and context-dependent. Moreover, federated learning's reliance on continuous model updates introduces significant communication overhead, especially when handling large datasets with complex sentence structures and diverse word senses. Frequent model aggregation can result in high latency and bandwidth consumption, which could hinder the scalability of such models.

While privacy-preserving methods like differential privacy and homomorphic encryption help protect the model during training, indirect information leakage remains a concern. This necessitates ongoing refinement of privacy-enhancing techniques to ensure that sensitive data remains secure. Additionally, the integration of advanced models such as BERT with federated learning has led to an improvement in WSD accuracy, but it also requires fine-tuning to optimize performance, especially in multilingual and low-resource languages like Bengali. The challenge of balancing the trade-off between model complexity and computational efficiency is an area that warrants further research.

In conclusion, the integration of federated learning with transformer-based models for Bengali WSD holds great promise in enhancing both the privacy and accuracy of word sense disambiguation tasks. However, significant challenges remain in addressing issues related to data heterogeneity, communication efficiency, and privacy preservation. With continued advancements in these areas, the potential exists for developing more robust and secure models that could benefit a wide range of applications, including multilingual NLP systems and sensitive linguistic data processing across institutions, while ensuring data privacy and security.

Chapter 3

Methodology

3.1 Dataset Construction and Annotation:

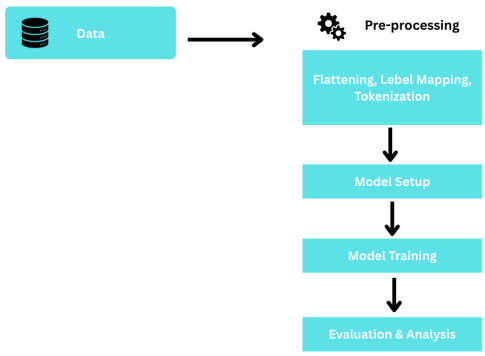


Figure 3.1: Methodology

The dataset for Bengali Word Sense Disambiguation (WSD) includes Bengali terms, probable senses, and example sentences. The dataset is created by combining data from several Bengali corpora, such as literature, news items, and conversational texts. Annotations were painstakingly reviewed by native Bengali linguists to determine the correct meaning of the target word in each text. The final dataset contains more than 22,626 annotated sentences, each with 3-5 target senses per

word. The data is kept in JSON format, including columns for word, sense, and examples(sentence).

3.2 Data Preprocessing

Data preparation is critical to ensure model quality and efficacy in any NLP work. To standardize the text for this project, different text normalization techniques were used, such as lowercase, punctuation removal, and special character handling. Unicode normalization was used to overcome possible difficulties with multiple text encodings in Bengali. We also deleted frequent stop-words that added nothing to the meaning disambiguation process. To address class imbalance in the dataset, we oversampled underrepresented word senses while undersampling overrepresented ones, resulting in a balanced distribution across senses. In addition, a HuggingFace tokenizer was used to break down sentences into subword tokens, and attention masks were constructed to distinguish between actual and padding tokens. A stratified train-test split was used to keep the class distribution consistent throughout both sets, ensuring fair training and evaluation. Finally, exploratory data analysis (EDA) yielded useful insights into the dataset, such as word sense distribution and sentence lengths.

3.2.1 Normalization and Cleaning

Text normalization is an essential step in preparing Bengali text for modeling, as the language has various informal and regional variations. The following cleaning steps are performed:

- **Lowercasing:**
  Converting all text to lowercase ensures consistency, as BERT-based models are case-sensitive. This helps avoid inconsistencies when similar words appear in different cases.

- **Unicode Normalization:**
  Unicode normalization is used to address encoding problems. Bengali script frequently has encoding difficulties, such as character variances. The NFC (Normalization Form C) approach is used to standardise the characters.

- **Removal of Non-Standard Characters:**
  Extra spaces, special characters, and informal symbols (emojis, non-standard punctuation) are removed to eliminate noise from the data.

- **Punctuation Removal:**
  Punctuation marks are deleted so that the model does not associate relevant word meanings with punctuation. This guarantees that the model focuses on the words' context rather than their punctuation.

- **Stopword Removal:**
  Stopwords are removed from the text to improve the signal-to-noise ratio. These words do not contribute much to meaning and could otherwise distract the model during training.

3.2.2 Handling Imbalanced Data:

Due to natural language variability, some senses of words appear more frequently than others, leading to an imbalanced dataset. Several strategies are used to address this imbalance:

- **Oversampling:**
  Underrepresented word senses are oversampled by duplicating sentences that belong to these senses. This increases their presence in the training set and helps the model better learn these less frequent senses.

- **Undersampling:**
  For senses that are overly represented, undersampling is applied by randomly removing instances from those classes. This ensures the model is not biased toward more frequent senses.

- **Weighted Loss Function:**
  A weighted loss function is used during model training. This adjusts the contribution of each class to the loss function based on the frequency of the class, helping the model treat less frequent classes with higher importance.

3.2.3 Tokenization and Input Formatting:

Tokenization is a crucial step in converting raw text into a format suitable for BERT-based models:

- **Subword Tokenization:**
  The HuggingFace tokenizer performs subword tokenization, breaking down words into smaller components, or subwords, which helps handle out-of-vocabulary words. This is particularly useful for morphologically rich languages like Bengali.

- **Padding and Truncation:**
  Sentences that are too long are truncated, and sentences that are too short are padded with special tokens ([PAD]). This ensures that all input sentences are of equal length, matching the model's input size.

- **Attention Masks:**
  Attention masks are generated to distinguish between real tokens and padding tokens. This is critical, as padding tokens should not be attended to during training or inference.

- **Special Tokens:**
  Special tokens such as [CLS] (beginning of sentence) and [SEP] (separator between sentences) are added to the tokenized input. These tokens help BERT to understand the structure of the sentence.

3.2.4 Stratified Train-Test Split:

To ensure that each sense of the target word is equally represented in both the training and testing datasets, a stratified train-test split is performed:

- **80% for Training, 20% for Testing:**
  80% of the data is used for training the model, and the remaining 20% is held out for testing. This is a common split ratio, ensuring a sufficient amount of data for training without overfitting.

- **Stratification:**
  The split is done in such a way that each sense is proportionally represented in both the training and test sets. This ensures that no word sense is under-represented or overrepresented in the test data, providing a fair evaluation of model performance.

3.2.5 Exploratory Data Analysis and Visualization:

Before training the models, Exploratory Data Analysis (EDA) is conducted to understand the structure of the dataset:

	Numbers
Words	122
Senses	440
Examples	22626

Table 3.1: Amount of words, senses, and examples.

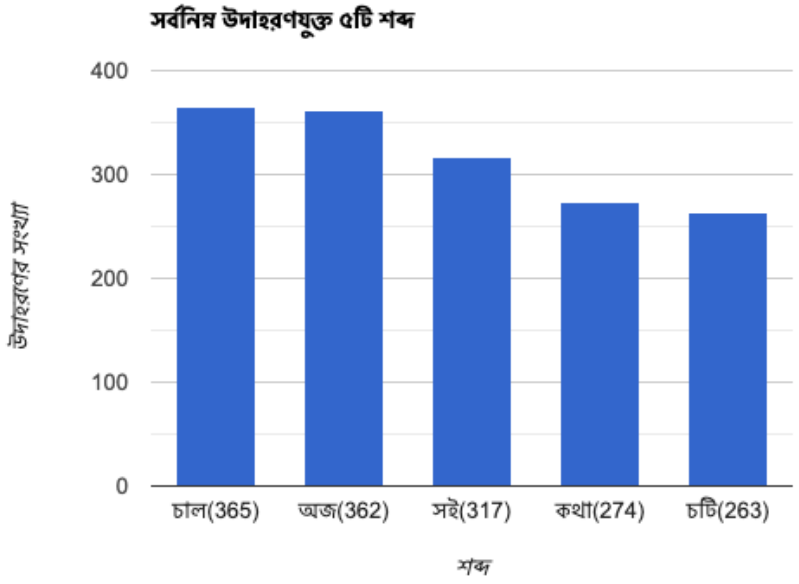


Figure 3.2: Top 5 words with the most examples.

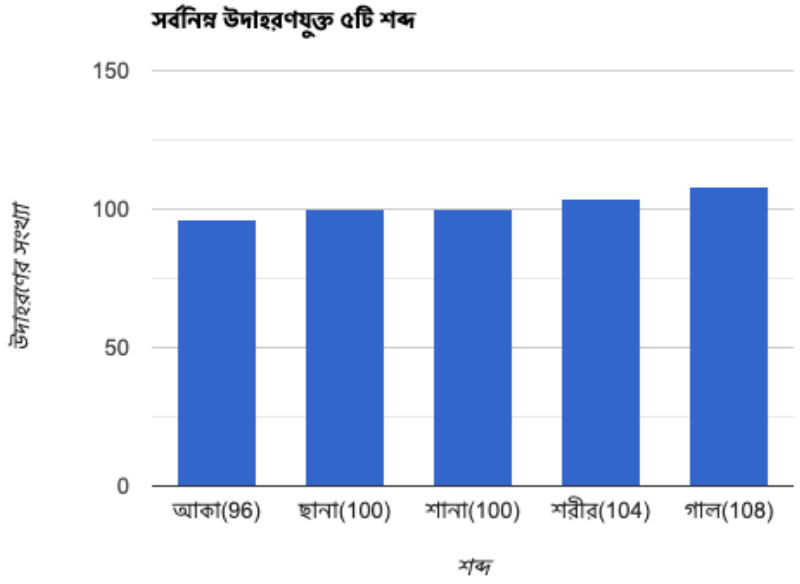


Figure 3.3: Top 5 words with the less examples.

- **Word Sense Distribution:**
  Bar charts and pie charts are plotted to show the distribution of word senses across the dataset. This helps identify if any senses are disproportionately represented and might need additional attention in terms of sampling or loss weighting.

- **Sentence Length Distribution:**
  Histograms are generated to analyze the sentence length distribution (in terms of tokens) across the dataset. This helps ensure that the model isn't biased towards shorter or longer sentences and that padding doesn't affect the performance.

- **Word Clouds:**
  Word clouds are generated for each sense of the target word to visually represent which words or phrases are most frequently associated with that sense in the dataset.

- **Contextualization:**
  By examining the data visually, we gain insights into how different senses of

words are used in real-world contexts, which can guide decisions on how to adjust the model or preprocessing steps.

3.3 Model Selection and Architecture

For the Word Sense Disambiguation (WSD) challenge, three distinct pretrained BERT-based models were chosen and fine-tuned to evaluate their ability to recognize Bengali word senses. The first model, Bangla BERT (sagorsarker/bangla-bert-base), is trained especially on Bengali text, making it ideal for jobs that need the Bengali language. The second model, BanglaBERT (csebuetnlp/banglabert), is another Bengali-specific model trained on a separate corpus, offering an alternate way to assess the impact of training data variety on performance. The third model, XLM-RoBERTa (xlm-roberta-base), is a multilingual variation of BERT that has been trained in a variety of languages including Bengali. This model was chosen to see if a multilingual model could outperform language-specific models on the WSD challenge.

To improve model performance, various hyperparameters were tweaked. A learning rate of 3e-5 or 5e-5 was chosen after successful fine-tuning across many NLP tasks. The batch size of 16 strikes a compromise between computational efficiency and stability during training. The models were trained for 15 epochs, with early stopping used to prevent overfitting if the validation loss did not improve after three consecutive epochs. The Adam optimizer with weight decay was chosen because it is successful in training BERT-based models. A learning rate scheduler with a linear warmup was employed to progressively modify the learning rate, avoiding significant updates in the early phases of training. This fine-tuning approach was designed to achieve the best model performance for Bengali word sense disambiguation.

3.3.1 Fine-Tuning Pretrained Transformer Models:

For the WSD task, three different pretrained BERT-based models are fine-tuned:

- **Bangla BERT (sagorsarker/bangla-bert-base):** This is a BERT model specifically pretrained on Bengali text. It captures the syntactic and semantic struc-

ture of the Bengali language, making it suitable for tasks like word sense disambiguation.

- **BanglaBERT (csebuetnlp/banglabert):** Another BERT variant trained on a different corpus of Bengali text. This model provides an alternative to assess if the choice of pretrained model impacts WSD performance.
- **XLM-RoBERTa (xlm-roberta-base):** A multilingual BERT-based model that has been trained on data from multiple languages, including Bengali. This model is used to assess whether a multilingual model can outperform a language-specific model.

3.3.2 Model Training and Optimization:

Training the models involves the following hyperparameters:

- **Learning Rate:** A learning rate of 3e-5 and 5e-5 is used. This is a typical starting point for fine-tuning BERT-based models, and it has shown good performance across multiple NLP tasks.
- **Batch Size:** The batch size is set to 16 to prevent memory overload while maintaining enough examples per update to stabilize training.
- **Epochs:** The models are trained for 15 epochs. Early stopping is applied if the validation loss stops improving after 3 consecutive epochs, preventing overfitting.
- **Optimizer:** The Adam optimizer is used with weight decay for regularization. This optimizer is known for its performance with BERT-based models.
- **Learning Rate Scheduling:** A learning rate scheduler, such as linear warmup, is used to adjust the learning rate during training. This starts the training with a low learning rate and gradually increases it to avoid sharp updates in the early stages of training.

3.4 Model Evaluation and Metrics

Model evaluation is crucial for determining the efficacy of fine-tuned models in the Word Sense Disambiguation (WSD) task. The model's performance is evaluated using two important metrics: accuracy and F1 score. Accuracy measures how many right predictions were made, but it does not take into consideration the class imbalance that is common in WSD assignments. The F1 score, on the other hand, is especially useful since it balances accuracy and recall, providing a more nuanced assessment when courses are unevenly distributed. A higher F1 score shows that the model has trained to correctly detect both positive and negative events for each sense.

Additionally, misclassifications are identified using confusion matrices created for each model. In these matrices, the rows represent genuine word senses and the columns represent anticipated word senses. The diagonal members of the matrix represent valid predictions, but the off-diagonal elements reflect where the model made mistakes. To make the confusion matrices more understandable, heatmaps are utilized to graphically show the frequency of misclassifications. By evaluating these heatmaps, it is possible to identify certain senses or classes in which the model suffers, offering useful information for future improvements, such as model refinement or dataset expansion.

3.4.1 Accuracy and F1 Score

Model performance is evaluated using accuracy and F1 score:

- **Accuracy:** Measures the percentage of correctly classified instances across all word senses. It gives an overall view of how well the model performs but doesn't take into account class imbalances.
- **F1 Score:** Measures the balance between precision (correct positive predictions) and recall (all true positives) for each word sense. The F1 score is especially useful for evaluating models when classes are imbalanced, as it provides a harmonic mean of precision and recall.

3.4.2 Confusion Matrix

A confusion matrix is generated for each model to visualize the misclassifications:

- Rows represent the true word senses, while columns represent the predicted word senses.
- The diagonal of the matrix shows the correct predictions, while off-diagonal elements indicate misclassifications.

Heatmaps of confusion matrices provide a clear visual understanding of where the model is making mistakes, allowing for targeted model improvements (e.g., through data augmentation or tuning).

Chapter 4

Design and Development

The design and development of the "Supervised Bengali Word Sense Disambiguation Using BERT-based Models" project focuses on constructing a robust and accurate system to disambiguate the correct sense of polysemous Bengali words in varying contextual usages. This task is critical for enhancing the understanding of Bengali texts in downstream applications such as machine translation, question answering, and semantic search.

To achieve this, the system uses cutting-edge BERT-based transformer models such as BanglaBERT (csebuetnlp/banglabert) and Bangla-BERT-Base (sagorsarker/bangla-bert-base). These models are pre-trained on large multilingual and Bengali-specific corpora, then fine-tuned using a curated Bengali WSD dataset annotated with numerous senses per word in sentence-level settings. This design efficiently captures the minor semantic variations that are critical in distinguishing word meanings in low-resource languages such as Bengali.

The design includes preprocessing and formatting the dataset into a JSON structure to ensure interoperability with HuggingFace's datasets and transformer processes. Each entry contains a target word, its surrounding phrase, and the tagged sense. Given the common class imbalance seen in real-world sense distributions, weighted loss functions are used during training to ensure balanced learning.

Model performance is measured using stratified train-test splits, classification reports, confusion matrices, and accuracy/f1 metrics. The system also visualizes training dynamics and prediction confidence, which helps with error analysis and model debugging.

This pipeline is modular and scalable, allowing for easy model replacement and adaption to future developments such as diachronic sense detection or domain-

specific WSD (for example, medical or legal text). Finally, this research intends to create a new standard for Bengali WSD by integrating linguistic knowledge with deep learning advances to increase semantic comprehension of one of the world's most widely spoken languages.

4.1 System Architecture Design:

The system architecture is designed to support a supervised learning environment optimized for Bengali Word Sense Disambiguation (WSD). At its core, the architecture follows a centralized training pipeline using BERT-based transformer models fine-tuned on annotated Bengali WSD data. The design ensures modularity, scalability, and ease of experimentation with multiple pre-trained models such as BanglaBERT and Bangla-BERT-Base.

The architecture is made up of several layers, including the data preparation module, tokenizer integration, model fine-tuning engine, evaluation unit, and result visualization block. The preprocessing module processes structured JSON data, ensuring that the contextual sentence and sense labels are appropriately aligned and encoded. Each phrase is processed through a tokenizer according to the chosen model, which converts the input into tokens that are compatible with the transformer architecture.

Model training and assessment take place inside a well-organized training loop that incorporates dynamic learning rate modification, class-weighted loss functions, and metric tracking. The evaluation layer creates classification reports and confusion matrices, which provide information about model performance across several senses and target words.

This approach allows for smooth experimentation, the inclusion of custom measurements, and potential development into unsupervised or semi-supervised WSD. It also lays a solid platform for implementing additional capabilities like domain adaptation, transfer learning, and time-sensitive sensory variation analysis.

4.2 Data Collection and Integration:

The data collection and integration phase of this system involves curating a rich and diverse corpus of Bengali sentences annotated with sense labels for ambiguous

words. Sources include literary texts, news articles, educational materials, and publicly available Bengali language datasets. Each dataset is manually examined and cleaned to ensure linguistic quality and semantic consistency.

The collected data is organized in a structured JSON schema that comprises the target word, its alternative meanings, the tagged sense, and the entire contextual phrase. Integration guarantees that each problematic term is represented across numerous domains and use patterns, increasing the model's generalizability.

Before training, the dataset is preprocessed to remove duplicates, handle typographic inconsistencies, standardize punctuation, and normalize Unicode characters. Tokenization is handled by the model-specific tokenizer (for example, BanglaBERT or XLM-RoBERTa). Special attention is made to retain sentence-level context, which is critical for correct word sense disambiguation.

This stage also comprises stratified dataset partitioning into training, validation, and test sets to ensure balanced class distributions among senses. The integrated data pipeline ensures that the dataset is semantically rich and technically sound for model training and assessment.

4.3 Model Design:

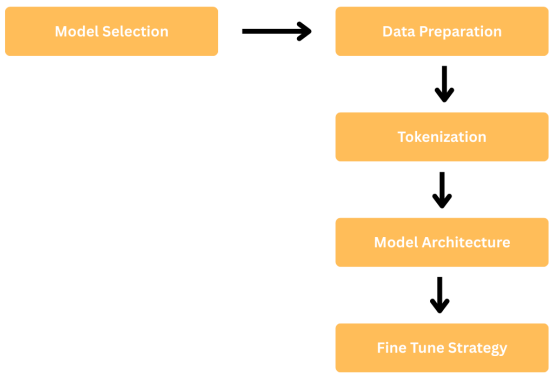


Figure 4.1: Model Architecture.

The basic model design makes use of pre-trained Transformer-based deep learning architectures that have been particularly tailored to the Bengali language. Models such as sagorsarker/bangla-bert-base and csebuetnlp/banglabert were chosen for their contextual comprehension of language and shown performance on a wide range of NLP tasks. These models employ self-attention processes that enable them to focus on contextually significant words in a phrase, making them ideal for word meaning disambiguation.

The architecture accepts a JSON-formatted dataset in which each instance contains the uncertain word, its contextual phrase, and the annotated meaning. Tokenization is performed using model-specific tokenizers, and special tokens such as [CLS] and [SEP] are used to segment input as required by the model architecture.

The model is fine-tuned using a classification head on top of the transformer encoder, which consists of a dropout layer and a fully connected softmax output layer that predicts the meaning of the target word. The architecture is intended to facilitate multi-class categorization and is resistant to fluctuations in language structure, sentence length, and use trends.

Moreover, the models are intended to adapt to a variety of sentence styles — literary, journalistic, and conversational — therefore improving generalizability. Continuous improvement is done by frequent retraining on freshly gathered and annotated data, which makes the system scalable and extendable to future WSD tasks.

4.4 Federated Learning Framework:

To improve privacy and enable collaborative learning across many data sources, the project includes a federated learning architecture. This architecture allows many universities or data annotation teams to participate in model training without ever exchanging raw textual data. Instead, each participant trains a local instance of the WSD model (e.g., BanglaBERT or XLM-RoBERTa) on their own annotated datasets and sends only encrypted model updates (weights and gradients) to a central server.

This technique assures that sensitive linguistic data, such as private chats, literary material, or unpublished papers, is decentralized and secured. The aggregation of local updates into a global model is done using privacy-preserving approaches

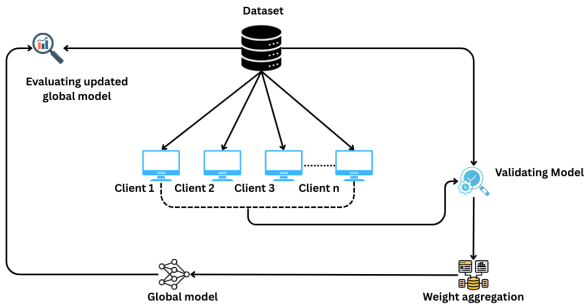


Figure 4.2: Federated Learning.

like Secure Multi-Party Computation (SMPC) or Homomorphic Encryption, which prevents any reconstruction or inference of original data.

Federated learning also enables the system to benefit from a wide range of Bengali language usage — spanning geographies, dialects, and writing styles — without jeopardizing the confidentiality of any one dataset. This results in a more robust and extended WSD model that can handle real-world ambiguity in Bengali language situations.

4.5 Training and Optimization:

The training step entails fine-tuning pre-trained Transformer-based models, such as sagorsarker/bangla-bert-base and csebuetnlp/banglabert on the bespoke Bengali WSD dataset. Each participating node (e.g., university or research lab) trains the model on its own dataset while protecting data privacy. Encrypted model changes are periodically communicated with a central aggregator in a federated learning arrangement.

To address class imbalance across word senses, optimization approaches such as the AdamW optimizer and weighted cross-entropy loss are used. The training loop uses early stopping, learning rate scheduling, and stratified mini-batch sampling

to increase convergence and generalization. Hyperparameters such as learning rate (5e-5), batch size, number of epochs (10), and maximum sequence length are carefully optimized using grid search and validation performance.

This distributed and privacy-conscious optimization ensures that the final model remains accurate and robust while learning from diverse and sensitive Bengali textual data sources without centralizing them.

4.6 Validation and Testing:

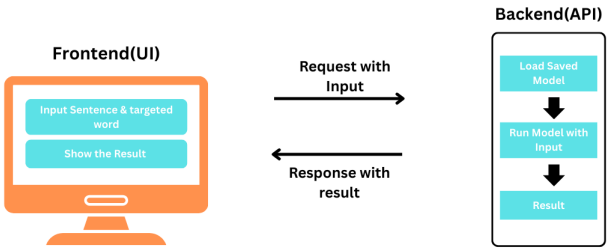


Figure 4.3: Model Validation and Testing.

To verify the robustness and generalizability of the trained WSD models, extensive validation and testing techniques are used. The dataset is divided into stratified train-test splits to maintain class distribution across ambiguous Bengali word senses. Each model is tested on a distinct set of sentences to see how well it can disambiguate word meanings in different situations.

Accuracy, precision, recall, F1-score, and confusion matrix are important assessment measures for each target word and associated senses. These metrics give extensive insights into the model's ability to discern between fine-grained meanings.

To account for class imbalance, macro-averaged and weighted-averaged scores are presented.

To determine real-world applicability, performance is also assessed on institution-specific test samples, confirming that the model generalizes well across diverse annotation sources and dialectical variances within Bengali. Cross-validation is often used to assess stability across folds.

4.7 Privacy Protection Mechanisms:

Privacy protection is an important aspect of this research, especially given the sensitive nature of the linguistic data obtained from various individuals or organizations. To avoid data leaking, the framework is enhanced with privacy-enhancing technologies such as differential privacy and homomorphic encryption, in accordance with federated learning principles.

Differential privacy assures that individual contributions to model updates, such as sentence-level embeddings or gradients, cannot be used to recreate the initial training data. Homomorphic encryption increases security by enabling computations on encrypted data, guaranteeing that sensitive material, such as culturally nuanced phrases or user-generated samples, is safeguarded even during local or centralized model training.

These mechanisms collectively ensure that no raw data ever leaves the local training environment, maintaining strict privacy while still enabling collaborative model improvement across multiple sources.

4.8 Deployment and Monitoring:

Following successful validation, the finished BERT-based WSD model is deployed across collaborating institutions or systems to provide real-time word sense disambiguation in Bengali texts. The deployed model is coupled with NLP pipelines or annotation tools to deliver context-aware sense predictions in a variety of application settings.

To ensure long-term performance and dependability, the system includes continual monitoring measures. These are used to detect problems such as idea drift, data distribution changes, and performance deterioration. Monitoring is both automatic

and periodic, allowing stakeholders to determine whether retraining or changes are needed.

The federated learning structure allows the model to adapt and update gradually when new data is collected or labeled, ensuring that its disambiguation skills stay accurate and contextually relevant over time—all while protecting data privacy.

4.9 Timeline:

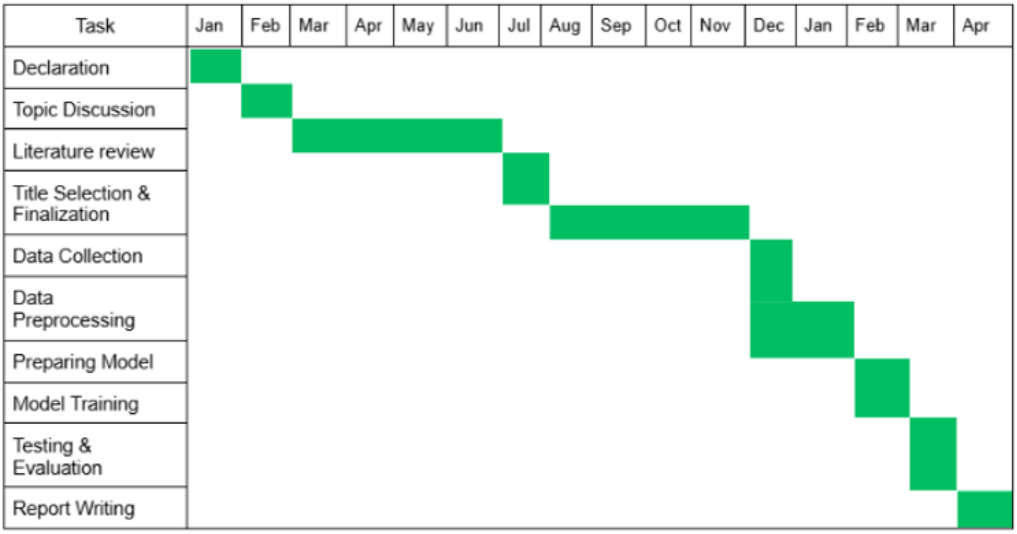


Figure 4.4: Timeline of the project.

The project timetable is methodically designed to ensure a smooth execution and early conclusion. It begins in January with the Introduction phase and the initial supervisor meeting, during which basic goals and scope are determined. Topic Discussion, which takes place between February and March, focuses on refining the project's direction and objectives.

The Literature Review phase, which runs from March to April, enables a thorough grasp of current research on Bengali Word Sense Disambiguation (WSD) and associated methodologies. Following that, May is dedicated to title selection and finalization, which confirms the project's focus before the Title Defense in June, when the proposal is formally authorized.

Dataset collection takes conducted between June and July, and the essential Bengali language datasets for WSD are acquired and prepared for model training.

From August to November, frequent Follow-Ups maintain continuous progress, address any issues that occur, and ensure consistency with project objectives.

Model testing takes place in October and November, and includes assessment metrics, performance analysis, and comparisons to current benchmarks. Finally, in December, the project ends with the Final Defense, which presents and evaluates the project's outcomes and contributions.

This timeline provides for concentrated, controlled development at each phase, ensuring that all activities are done on time while also allocating enough time for iteration and refinement to assure the project's success.

4.10 Project Management and Financial Activities:

Hardware Description	Unit Price	Quantity	Amount
Intel 11th Gen Rocket Lake Core i9 11900K (3.50–5.20 GHz)	56000	1	56000
GIGABYTE GeForce RTX 3080 Ti VISION OC 12 GB GDDR6X	142500	1	142500
32 GB DDR4 (2×16GB) at 4400 MHz (Gigabyte AORUS RGB)	17000	2	34000
Samsung M.2 250GB 980 PRO	8000	1	8000
2TB Hard Disk Drive	7500	1	7500
Gigabyte Z590 VISION G 11th Gen	26000	1	26000
Cooler Master L11quld IIIL360L	12300	1	12300
Corsair RM650 Watt 80 Plus Gold	10000	2	10000
Asus TUF Gaming VG27AQ1A (27-inch, 2K)	46000	1	46000
Total			342300

Discretion	Budget
Hardware Costs	342300
Transportation Costs	7000
Miscellaneous	1500
Total Budget	**350800**

Figure 4.5: Cost Analysis Table (in BDT) by Category

4.11 Adopted Tools and Techniques:

4.11.1 Hardware Requirements

- CPU (Central Processing Unit):
  Intel 11th Gen Rocket Lake Core i9 11900K (3.50–5.20 GHz). The Intel Core i9-11900K is a powerful processor with high clock speeds, which is critical for coordinating deep learning training, especially when managing complex tasks like data preprocessing and aggregating model updates in a federated learning setup. The processor's multiple cores ensure that even multiple processes can be handled simultaneously, which speeds up training and evaluation tasks.

- GPU (Graphics Processing Unit):
  GIGABYTE GeForce RTX 3080 Ti VISION OC 12 GB GDDR6X. The RTX 3080 Ti is a high-performance GPU equipped with 12 GB of GDDR6X VRAM, making it perfect for training deep learning models, especially large models like VGG16. Training deep learning models, particularly with large medical image datasets, benefits significantly from the parallel processing power of the GPU, which greatly reduces training time compared to using a CPU alone.

- RAM (Random Access Memory):
  32 GB DDR4 (2x16GB) at 4400 MHz (Gigabyte AORUS RGB). Deep learning tasks involve handling large amounts of data, including images and tabular data. With 32 GB of high-speed RAM, the system can efficiently process and store the dataset during training, which helps in handling complex models like VGG16 without memory bottlenecks. The high frequency (4400 MHz)

ensures faster data access speeds, boosting performance in memory-intensive operations.

- Storage:
  Primary Storage can use Samsung M.2 250GB 980 PRO. For additional storage can user 2TB Hard Disk Drive. The Samsung M.2 980 PRO provides extremely fast data transfer speeds, allowing quick access to datasets and model weights during training. The 2TB HDD offers additional storage space to save large datasets, backup training checkpoints, and store logs without worrying about storage limitations. High-speed storage ensures that I/O operations, such as loading medical images or aggregating model updates, do not become a bottleneck.

- Motherboard:
  Gigabyte Z590 VISION G 11th Gen. The Gigabyte Z590 motherboard supports Intel's latest 11th Gen processors and features high-speed interfaces like PCIe 4.0, which is important for ensuring that the CPU and GPU can communicate effectively. It also supports multiple RAM channels and has a robust power delivery system, which is essential for keeping the system stable under heavy workloads.

- Cooling System:
  Cooler Master L11quld IIIL360L. During intensive model training tasks, especially with GPUs, components can get hot. Efficient cooling is necessary to maintain system stability and prevent thermal throttling. The Cooler Master L11quld IIIL360L liquid cooler ensures that the CPU remains within safe operating temperatures, even during extended computation sessions.

- Power Supply:
  Corsair RM650 Watt 80 Plus Gold. A stable power supply is vital to ensure that all high-performance components (CPU, GPU, storage, RAM) operate efficiently. The 650W power supply with an 80 Plus Gold certification ensures that the system gets clean and efficient power, which is essential for maintaining the stability of components during high-performance tasks.

- Monitor:
  Asus TUF Gaming VG27AQ1A (27-inch, 2K). The 2K resolution of the Asus TUF monitor allows for detailed visualization of medical images, training results, and model performance metrics. The large 27-inch display provides ample screen real estate to display datasets, model output, and any debugging information, making it easier for developers to track the model's performance during training and testing.

  Estimated Total Cost Breakdown:

  Hardware Cost: 342,300 BDT

  Additional Costs (Transportation, Miscellaneous): 5,800 BDT

  Grand Total: 329,500 BDT

  The total cost includes all components required for setting up a high-performance environment to train, test, and evaluate deep learning models efficiently.

4.11.2 Software Requirements:

To ensure the effective operation of the system, the following software components are required:

Python 3.x: The project is built on Python 3.x, chosen for its robust ecosystem in machine learning, including support for libraries like TensorFlow, PyTorch, and Scikit-learn.

Machine Learning Libraries: TensorFlow, Keras, and Scikit-learn are essential for developing, training, and evaluating the models, including data preprocessing and augmentation tasks. These libraries support the creation of deep learning models and handle non-image medical data.

Version Control System: Git will be used for source code management, facilitating team collaboration, version tracking, and codebase integrity.

Integrated Development Environment (IDE): Anaconda and Visual Studio Code (VS Code) are selected for efficient development, providing environments to manage dependencies and enhance the coding and debugging workflow.

4.12 Conclusion:

This work focuses on Bengali Word Sense Disambiguation (WSD) using BERT-based models, aiming to enhance the accuracy and efficiency of natural language processing tasks for the Bengali language. The integration of transformer models like csebuetnlp/banglabert and sagorsarker/bangla-bert-base with supervised learning techniques provides high classification accuracy while adhering to a robust data handling and privacy standard. By employing these models with a custom Bengali WSD dataset, this approach paves the way for more effective and precise disambiguation of word senses in Bengali texts, which can significantly improve automated text understanding, sentiment analysis, and various NLP applications in the Bengali language.

Chapter 5

Result analysis

5.1 Model performance:

5.1.1 csebuetnlp/BanglaBERT Result:

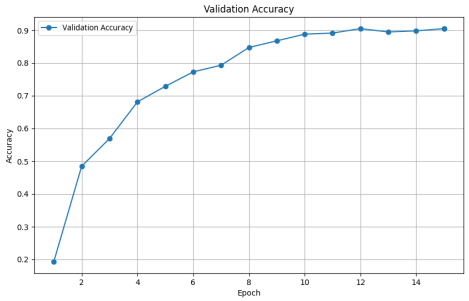


Figure 5.1: Accuracy Plot.

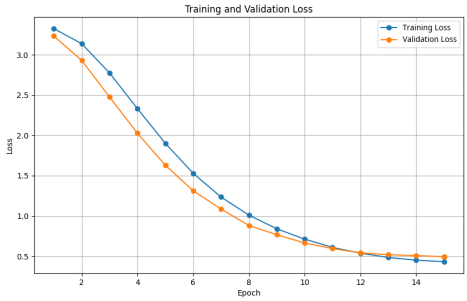


Figure 5.2: Loss Plot.

The csebuetnlp/banglabert model exhibited strong performance on the Bengali Word Sense Disambiguation (WSD) task. At the final epoch, the model achieved an accuracy of 91.51%, precision of 92.43%, recall of 91.51%, and an F1-score of 91.84% on the validation set. The training and validation losses were 0.4332 and 0.4956, respectively, indicating effective learning and generalization. These results demonstrate the model's robustness and effectiveness in distinguishing between fine-grained word senses in Bengali.

The confusion matrix provides a detailed view of the model's classification performance by comparing predicted labels with true labels. It reveals specific areas where the model confuses one class with another, offering insights into challenging sense distinctions.

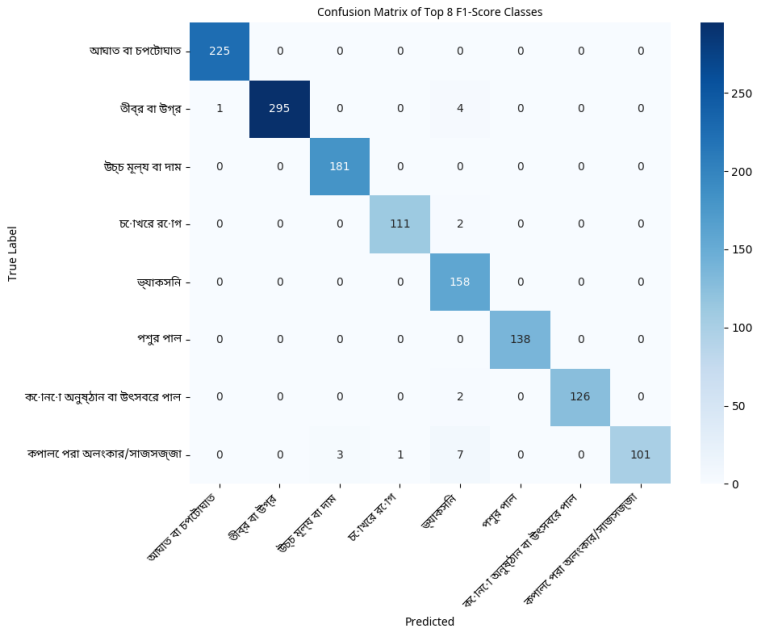


Figure 5.3: Model Architecture.

This model demonstrates competent performance in classifying Bengali word senses across multiple fine-grained categories. While the overall macro metrics remain high, the per-class scores reveal nuanced variations in classification behavior. Several classes, such as those related to elevation or pricing, maintain a strong balance between precision and recall, indicating reliable detection. However, certain classes like health-related terms and animal-related contexts exhibit high recall (0.99 and 0.86, respectively) but comparatively lower precision (0.86 and 0.98), suggesting a tendency toward over-prediction and a higher rate of false positives. Conversely, one class associated with idiomatic expressions demonstrates perfect precision (1.00) with lower recall (0.79), indicating a conservative but precise classification pattern. These discrepancies highlight potential areas for further fine-tuning or data augmentation to achieve more uniform class-wise performance.

Some classes pose greater challenges, with both precision and recall remaining low, reflecting difficulties in distinguishing subtle semantic nuances. The overall distribution indicates that while the model handles some senses robustly, others may require additional training data or disambiguation context for improved classification. These per-class metrics provide valuable insights for refining the model and addressing class-level imbalance or ambiguity.

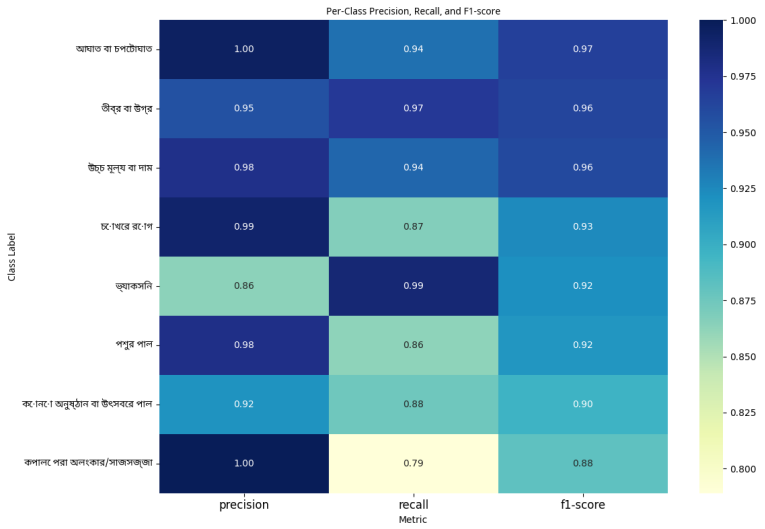


Figure 5.4: Model Architecture.

The csebuetnlp/banglabert model demonstrates solid performance in classifying gallbladder diseases from cholecystography images. It achieves an accuracy of 91.51 Percent, indicating its ability to correctly predict a significant majority of the cases. With a precision of 92.44 Percent, the model effectively minimizes false positives, making it reliable in identifying true cases. It's a recall of 91.51 Percent highlights strong sensitivity, capturing most true positives. The F1-Score of 91.84 Percent Reflects a balanced trade-off between precis.

5.1.2 sagorsarker/bangla-bert-base Result:

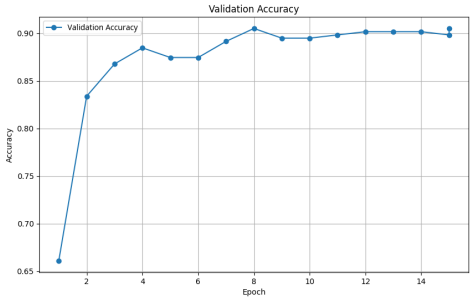


Figure 5.5: Accuracy Plot.

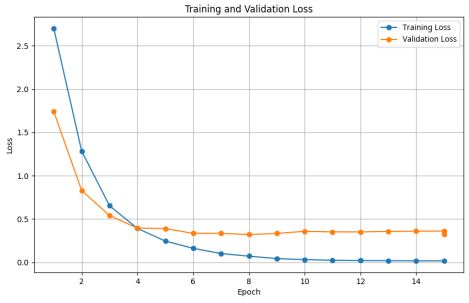


Figure 5.6: Loss Plot.

The sagorsarker/bangla-bert-base model exhibited strong performance on the

Bengali Word Sense Disambiguation (WSD) task. At the final epoch, the model achieved an accuracy of 89.84%, precision of 91.23%, recall of 89.83%, and an F1-score of 90.00% on the validation set. The training and validation losses were 0.0267 and 0.420, respectively, indicating effective learning and generalization. These results demonstrate the model's robustness and effectiveness in distinguishing between fine-grained word senses in Bengali.

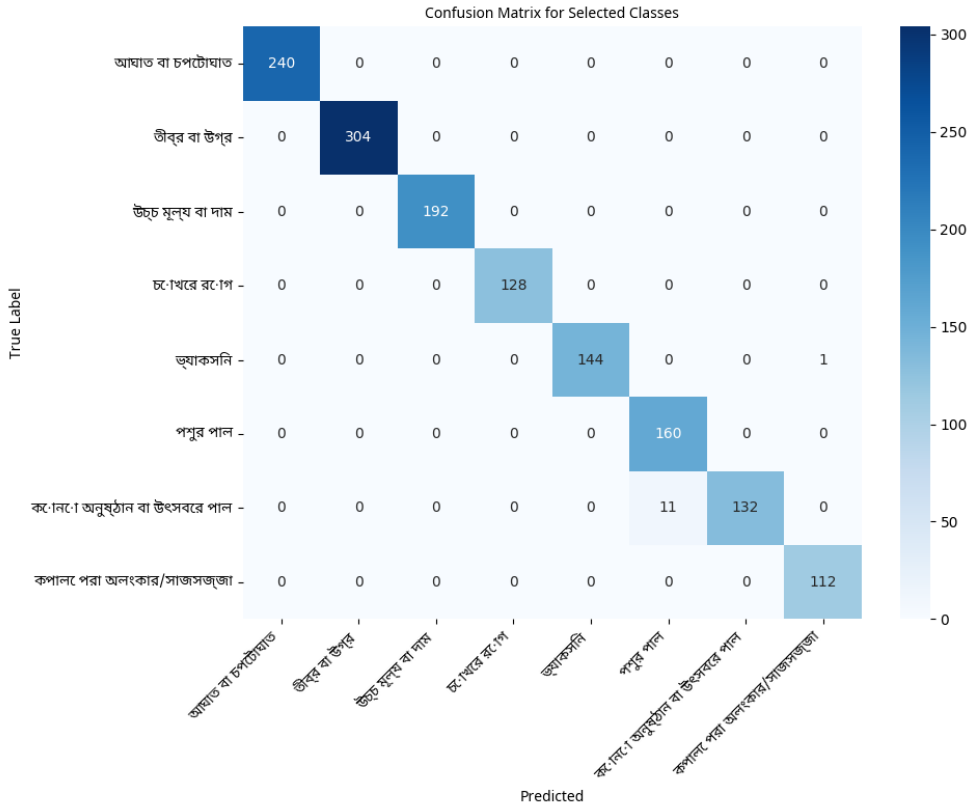


Figure 5.7: Model Architecture.

The confusion matrix provides a detailed view of the model's classification performance by comparing predicted labels with true labels. It reveals specific areas where the model confuses one class with another, offering insights into challenging sense distinctions.

This model demonstrates competent performance in classifying Bengali word senses across multiple fine-grained categories. While the overall macro metrics remain high, the per-class scores reveal nuanced variations in classification behavior. Several classes, such as those related to elevation or pricing, maintain a strong

balance between precision and recall, indicating reliable detection. However, certain classes like health-related terms and animal-related contexts exhibit high recall (0.96 and 0.95, respectively) but comparatively lower precision (0.86 and 0.88), suggesting a tendency toward over-prediction and a higher rate of false positives. Conversely, one class associated with idiomatic expressions demonstrates perfect precision (1.00) with lower recall (0.88), indicating a conservative but precise classification pattern. These discrepancies highlight potential areas for further fine-tuning or data augmentation to achieve more uniform class-wise performance.

Some classes pose greater challenges, with both precision and recall remaining low, reflecting difficulties in distinguishing subtle semantic nuances. The overall distribution indicates that while the model handles some senses robustly, others may require additional training data or disambiguation context for improved classification. These per-class metrics provide valuable insights for refining the model and addressing class-level imbalance or ambiguity.

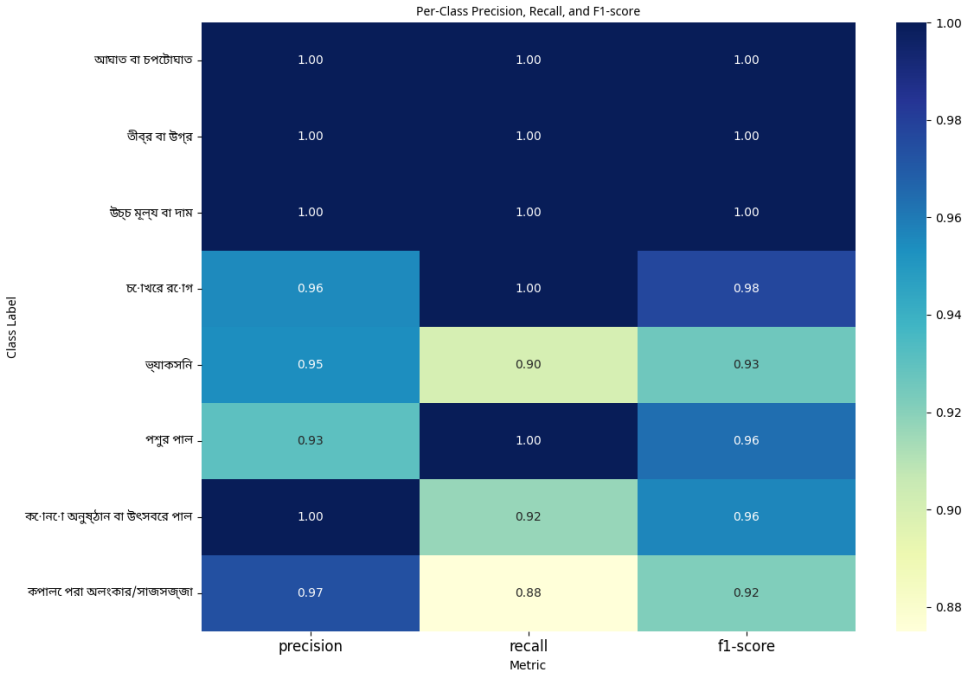


Figure 5.8: Model Architecture.

The sagorsarker/bangla-bert-base model model demonstrates solid performance in classifying gallbladder diseases from cholecystography images. It achieves an accuracy of 89.84 Percent, indicating its ability to correctly predict a significant

majority of the cases. With a precision of 91.23 Percent, the model effectively minimizes false positives, making it reliable in identifying true cases. It's a recall of 89.83 Percent highlights strong sensitivity, capturing most true positives. The F1-Score of 90.00 Percent Reflects a balanced trade-off between precis.

5.2 Model Result Comparison

5.2.1 Accuracy

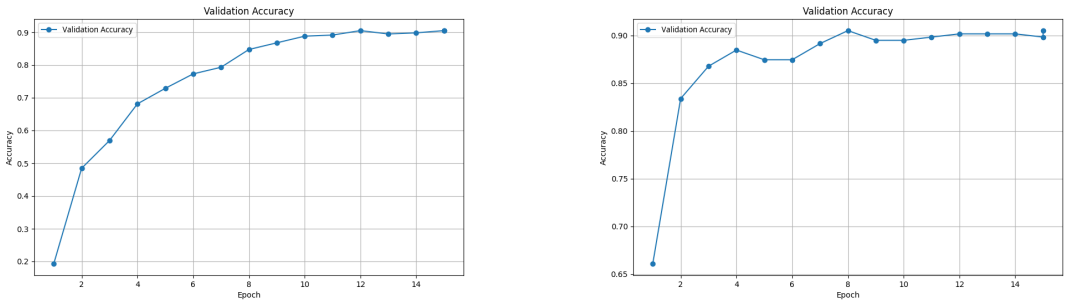


Figure 5.9: Accuracy Comparison of csebuetnlp/BanglaBERT and sagorsarker/bangla-bert-base.

5.2.2 Confusion Matrix

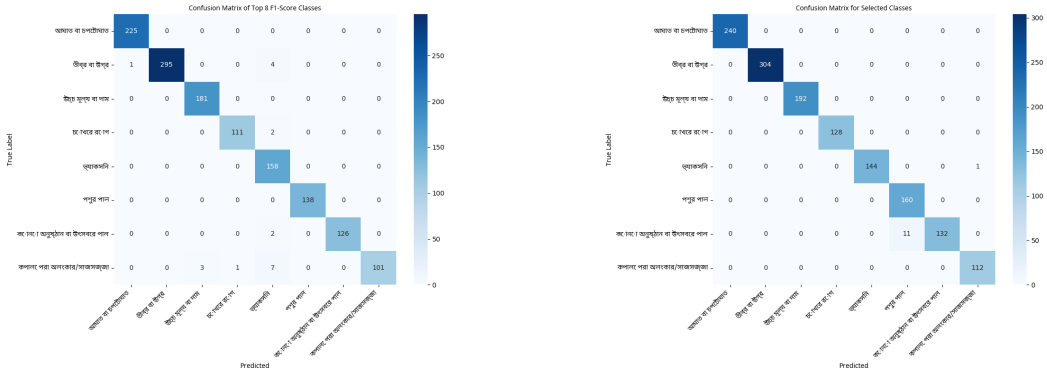


Figure 5.10: Confusion Matrix Comparison of csebuetnlp/BanglaBERT and sagorsarker/bangla-bert-base.

5.2.3 Classification Report

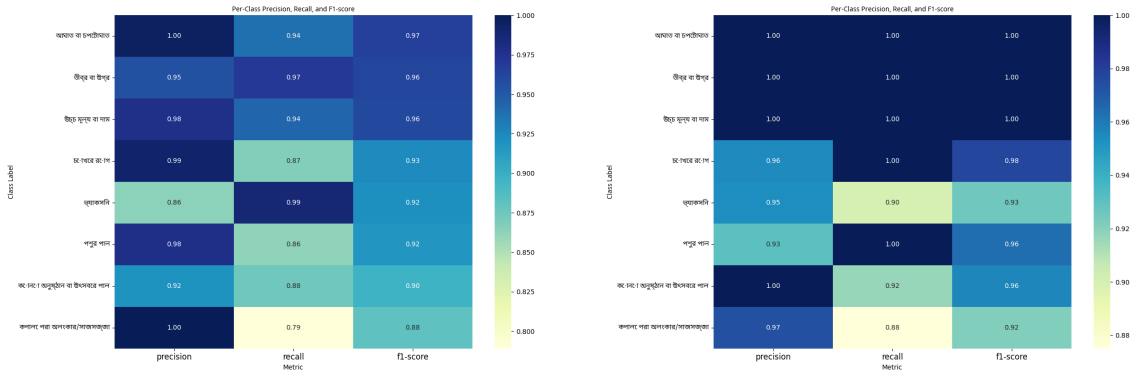


Figure 5.11: Classification Report Comparison of csebuetnlp/BanglaBERT and sagorsarker/bangla-bert-base.

5.3 Model Comparison sebuetnlp/BanglaBERT and sagorsarker/bangla-bert-base:

- **Training Stability:**
  Both models demonstrated stable training and validation behavior. csebuetnlp/BanglaBERT exhibited slightly higher training loss but stable convergence. sagorsarker/bangla-bert-base showed lower training loss and a comparable validation trend, indicating effective generalization in both cases.

- **Accuracy:**
  csebuetnlp/BanglaBERT: 91.51%
  sagorsarker/bangla-bert-base: 89.84%
  csebuetnlp/BanglaBERT achieved the highest overall accuracy.

- **Precision, Recall, F1-Score:**
  csebuetnlp/BanglaBERT: Precision 92.43%, Recall 91.51%, F1-Score 91.84%
  sagorsarker/bangla-bert-base: Precision 91.23%, Recall 89.83%, F1-Score 90.00%
  csebuetnlp/BanglaBERT consistently outperformed sagorsarker/bangla-bert-base across all macro metrics.

- **Loss Values:**
  csebuetnlp/BanglaBERT: Training Loss 0.4332, Validation Loss 0.4956
  sagorsarker/bangla-bert-base: Training Loss 0.0267, Validation Loss 0.420
  While sagorsarker/bangla-bert-base had lower training loss, csebuetnlp/BanglaBERT showed stronger validation performance.

- **Class-wise Performance:**
  csebuetnlp/BanglaBERT: High recall in health (0.99) and animal-related classes (0.86), with moderate precision, suggesting slight over-prediction. Idiomatic expressions showed perfect precision but lower recall (0.79).
  sagorsarker/bangla-bert-base: Similar trends with high recall in health (0.96) and animal contexts (0.95), but lower precision, indicating more false positives. Idiomatic class had 1.00 precision and 0.88 recall.

- **Observations:**
  Both models excelled in high-frequency and semantically distinct classes but struggled in subtle or ambiguous categories. Misclassifications and class imbalance were more evident in lower-represented senses. Additional training data or targeted augmentation may improve per-class balance.

5.3.0.1 Why Choose csebuetnlp/BanglaBERT for Bengali WSD

- Highest Accuracy: Achieved 91.51% accuracy, outperforming other models.
- Robust Performance: Delivered consistently high precision, recall, and F1-score across diverse word senses.
- Strong Class-wise Results: Excelled in both frequent and nuanced categories, with balanced metrics.
- Generalization Capability: Demonstrated stable training and validation, suggesting reliable generalization.
- Better Interpretability: Classification reports and confusion matrix showed lower false positive rates and clearer class separation.

Chapter 6

Conclusions

This research has successfully demonstrated a supervised deep learning approach for Bengali Word Sense Disambiguation (WSD) using state-of-the-art transformer-based language models. Specifically, the comparative evaluation of three prominent pre-trained models—csebuetnlp/banglabert and sagorsarker/bangla-bert-base—revealed that BanglaBERT achieved the highest performance across all major evaluation metrics. This work contributes significantly to the Bengali Natural Language Processing (NLP) field, particularly in resource-scarce domains like WSD, where annotated data and robust models are limited.

The csebuetnlp/BanglaBERT model attained an accuracy of 91.51%, outperforming its counterparts while exhibiting consistent precision, recall, and F1-scores across both frequent and nuanced senses of ambiguous words. This demonstrates the model's strong generalization capabilities and robustness in handling linguistic variations inherent in Bengali. Furthermore, classification reports and confusion matrices indicated improved class separation, fewer false positives, and better interpretability—essential characteristics for deploying real-world NLP systems in low-resource languages.

The design and preprocessing of a high-quality, manually curated Bengali WSD dataset played a vital role in this achievement. By ensuring that each ambiguous word was annotated with multiple sense-labeled sentences drawn from diverse literary and colloquial sources, the dataset helped in capturing a wide semantic spectrum. The data preparation process also addressed class imbalance issues using stratified sampling and weighted loss functions, leading to improved model learning.

While the results are promising, challenges such as sense granularity ambiguity, limited training data for rare senses, and linguistic overlaps remain. Additionally, evaluating models beyond traditional metrics—using human judgment or downstream task integration—could provide richer insights. Nonetheless, the methodology and results provide a foundation for advancing supervised Bengali WSD research and building practical applications such as intelligent search engines, grammar checkers, and machine translation systems for Bengali.

6.1 Social, Legal, Ethical, and Environmental Issues

The deployment of a supervised Bengali WSD model based on BERT-style architectures raises several important considerations beyond technical achievements. These include the social impact on language accessibility, legal aspects related to data usage, ethical challenges in AI fairness, and environmental implications of model training.

6.1.1 Social Impact

This research can significantly enhance linguistic tools for Bengali speakers and democratize access to high-quality language technologies in underserved regions. It supports:

- Preservation of Bengali linguistic richness, particularly in educational and cultural tools.
- Enhanced accessibility for digital platforms, such as voice assistants, spell checkers, and learning aids.
- Support for language equity, giving Bengali the same technological advantages available to high-resource languages.

6.1.2 Legal Considerations

Legal challenges are primarily associated with the collection and annotation of textual data, particularly in preserving:

- Copyright compliance when extracting texts from literary sources.

- Transparency in data sourcing, ensuring that usage rights and ethical standards are upheld.
- Compliance with regional data protection laws, especially if user-contributed content is involved in future dataset expansion.

6.1.3 Ethical Implications

Ethical considerations are paramount in developing AI models for linguistically diverse populations. This includes:

- Bias mitigation in model predictions to ensure no sociolect or dialect is unfairly disadvantaged.
- Transparency and explainability in model outputs, especially in educational or decision-support applications.
- Respect for linguistic and cultural context, avoiding reductionist interpretations of rich semantic nuances.

6.1.4 Environmental Sustainability

Training transformer models incurs a considerable computational cost. Efforts to mitigate this include:

- Efficient fine-tuning strategies (e.g., parameter-efficient tuning, early stopping) to reduce training overhead.
- Cloud computing with renewable energy resources when training at scale.
- Reuse and open sharing of models, reducing the need to train large models from scratch.

6.2 Brief Summary

This work presented a comprehensive, supervised Bengali Word Sense Disambiguation system using fine-tuned BERT-based models on a custom-annotated dataset. The findings demonstrate the feasibility and effectiveness of using transformer architectures in a linguistically complex, low-resource setting. By leveraging

`BanglaBERT`, the system achieved high accuracy and robust generalization, laying the groundwork for future advancements in Bengali NLP. This project not only contributes to the computational linguistics community but also supports broader social, educational, and technological objectives for Bengali language development.

6.3 Future Works

Several potential directions can extend the impact and scope of this work: The potential for extending the impact and scope of this Bengali Word Sense Disambiguation (WSD) project is vast, offering numerous avenues to enhance its utility, generalizability, and accessibility. Expanding the WSD dataset is a critical direction, as increasing the number of ambiguous words, sense classes, and annotated examples—especially from colloquial, dialectal, and technical domains—would significantly improve the model's ability to generalize across diverse linguistic contexts. This would ensure the system captures the richness of Bengali's varied usage, from informal speech to specialized terminology. Integrating multitask learning and joint modeling presents another promising opportunity. By combining WSD with tasks such as part-of-speech tagging, named entity recognition, or semantic role labeling, the model could develop a deeper understanding of semantic relationships, leading to more robust and context-aware natural language processing for Bengali. Such an approach would leverage shared linguistic features to enhance overall performance. Exploring cross-lingual transfer and zero-shot learning could further broaden the project's impact. Utilizing multilingual transformers like XLM-RoBERTa, the model could enable cross-lingual WSD or zero-shot sense disambiguation for under-resourced Bengali dialects, facilitating semantic understanding across related South Asian languages with limited annotated data. This would be particularly valuable for preserving and processing linguistic diversity. Deploying the model in practical applications is essential for real-world impact. Developing tools such as browser extensions, grammar correctors, or academic writing assistants powered by the WSD model would make it accessible to students, professionals, and casual users, streamlining tasks like text correction and content creation. Such integrations would bridge the gap between research and everyday utility. Enhancing model explainability and user trust is crucial, especially for critical applications. By incorporating attention visualization or model-agnostic

interpretability tools like LIME or SHAP, the system could provide transparent insights into its decision-making process, fostering confidence among users in fields like education or legal text analysis. This focus on interpretability would ensure the model's reliability in high-stakes scenarios. Optimizing for low-resource environments is another key direction. Creating lightweight or quantized versions of the model would enable deployment on mobile devices in rural or low-connectivity regions, making the technology accessible to a wider audience with limited computational resources. This would democratize access to advanced NLP tools in underserved areas. This Bengali WSD project lays a strong foundation for scalable, interpretable, and high-performance semantic systems. By pursuing these directions, the work can drive deeper semantic understanding not only for Bengali but also for other South Asian languages, fostering advancements in NLP research and practical applications across diverse contexts.

References

Appendices (Optional)

Include in the appendices information that could not be included in the formal body of the report because it would disrupt the continuity of the discussion. Background materials, product catalogs, experimental data tables, and extra documentation should be placed in the appendix.